

PYTHON ASSIGNMENT

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TOPIC: Enron Email Dataset

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Commands + Code + Text ▶ Run all Copy to Drive Connect ^
[ ] ▶ import pandas as pd
    !unzip "archive.zip"
    df = pd.read_csv('IMDB Dataset.csv')
    df.head()
... Archive: archive.zip
replace IMDB Dataset.csv? [y]es, [n]o, [A]ll, [N]one, [r]ename:
[ ] #Setup and Data Loading
    import pandas as pd
    import numpy as np
    df = pd.read_csv('IMDB Dataset.csv')
    df.head()
[ ] #Find the total number of reviews
    len(df)
[ ] #Find the distribution of sentiments
    df['sentiment'].value_counts()
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[ ] #Find the number of unique sentiments
    df['sentiment'].nunique()
[ ] #Convert sentiment to numeric (Positive=1, Negative=0)
    df['sentiment_score'] = df['sentiment'].apply(lambda x: 1 if x == 'positive' else 0)
    df.head()
[ ] #Find the average sentiment score
    df['sentiment_score'].mean()
[ ] #Find the length of each review
    df['review_length'] = df['review'].apply(len)
    df.head()
[ ] ▶ #Find the longest review in the dataset
    df.loc[df['review_length'].idxmax(), 'review']
[ ] #Find the shortest review in the dataset
    df.loc[df['review_length'].idxmin(), 'review']
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[1] #Find the shortest review in the dataset
df.loc[df['review_length'].idxmin(), 'review']
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[1] #Find the average length of a review
df['review_length'].mean()
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[1] #Find the median review length
df['review_length'].median()
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[1] #Standard deviation of review lengths
np.std(df['review_length'])
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[1] #Find reviews that contain the word "masterpiece"
df[df['review'].str.contains('masterpiece', case=False)]
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[1] #Count how many reviews are "positive"
len(df[df['sentiment'] == 'positive'])
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[1] #Find the top 5 longest positive reviews
df[df['sentiment'] == 'positive'].sort_values('review_length', ascending=False).head(5)
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[1] #Create a word count column for each review
df['word_count'] = df['review'].apply(lambda x: len(x.split()))
df.head()
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[1] #Find the average word count for positive vs negative reviews
df.groupby('sentiment')['word_count'].mean()
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[1] #Find reviews that are longer than 500 words
df[df['word_count'] > 500]
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[1] #Check for any missing (null) values in the dataset
df.isnull().sum()
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[1] #Extract the first 50 characters of each review
df['review_snippet'] = df['review'].str[:50]
df.head()
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[1] #Extract the first 50 characters of each review
df['review_snippet'] = df['review'].str[:50]
df.head()
```

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[1] #Export the processed dataframe to a new CSV
df.to_csv('IMDB_Processed.csv', index=False)
```